COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S1
DATE PERFORMED:	7/23/26	**DATE SUBMITTED:**	7/23/26
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ACTIVITY NO. _3__ Hands-on Activity 3.1 Linked Lists

1. PROCEDURE OUTPUT

HEADER FILE

dll > main.cpp > main()

```cpp
#include <iostream>
#include "doubly_LL.h"

int main() {
    doubleList<char>* head = newNode('C');

    // CREATION OF OTHER NODES
    doubleList<char>* second = newNode('P');
    doubleList<char>* third = newNode('E');

    // LINK THE NODES
    head->next = second;
    head->prev = nullptr;

    second->prev = head;
    second->next = third;

    third->prev = second;
    third->next = nullptr;

    // INITIAL TRAVERSAL
    std::cout << "Initial Traversal" << std::endl;
    dllTraverse(head);

    return 0;
}
```

MAIN FILE

```cpp
#ifndef DOUBLE_LL_H
#define DOUBLE_LL_H

#include <iostream>

template <typename T>
class doubleList {
public:
    T data;
    doubleList<T>* next = nullptr;
    doubleList<T>* prev = nullptr;

    doubleList(T val) : data(val), next(nullptr), prev(nullptr) {}
};

template <typename T>
doubleList<T>* newNode(T data) {
    return new doubleList<T>(data);
}

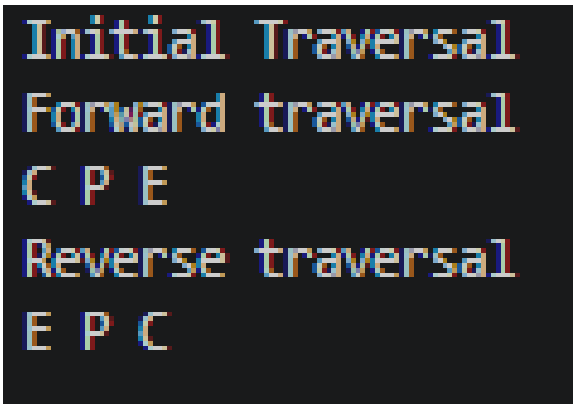
template <typename T>
void dllTraverse(doubleList<T>* currentNode) {
    doubleList<T>* tail = nullptr;

    std::cout << "Forward traversal \n";
    while (currentNode != nullptr) {
        std::cout << currentNode->data << " ";
        tail = currentNode;
        currentNode = currentNode->next;
    }
    std::cout << std::endl;

    std::cout << "Reverse traversal \n";
    while (tail != nullptr) {
        std::cout << tail->data << " ";
        tail = tail->prev;
    }
    std::cout << std::endl;
}

#endif
```

CODE OUTPUT



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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